

Abstract

The data set we investigated was Women's World Cup statistics from 1991-2019. This encompasses all eight tournaments that have been played, so it is actually population data, rather than a sample. There are 136 observations of 22 variables, including year, squad, average player age, number of matches played, goals per game, and many others. We also added three variables to the data set in order to answer our research questions. These are titled continent, win, and stage. The continent variable is a categorical variable with six levels, win is a binomial categorical variable, and stage is an ordinal categorical variable. We broke our data into categories using the continent variable and completed a Kruskal-Wallis test to determine if the average number of goals scored per game varies between continents. After completing the test using R, we found that there is a significant difference in the number of goals per game scored between Europe and Africa and Europe and Asia. Next, we created a logistic regression model to determine which variables are the best predictors of winning the World Cup and found that the most effective are goals per game and age. Finally, we expanded on this and performed an ordinal logistic regression to find out which variables are the best predictors of the stage of the tournament a team will make it to, and again, goals per game and age were the best predictors.

Question 1

Does the median number of goals scored per game differ between continents?

Introduction

To compare many means at the same time, one test that can be used is the ANOVA test. However, it requires certain conditions for normality, and unfortunately this data set does not meet them. Therefore, we must use a different test that does not require these conditions. One example is the Kruskal-Wallis test. It functions almost the exact same way as an ANOVA, it just allows for non-parametric data. Since it deals with skewed data, it compares the medians between groups, instead of the means, because the median is more resistant to outliers. This makes it a more reliable measure of center for this kind of data. To begin, we created a categorical variable with multiple levels. We chose to break the Women's World Cup data set down by continent, so there are levels for Africa, Asia, Europe, North America, Oceania, and South America. Then, we can compute the median for each of the six groups individually and determine if there is a statistically significant difference between medians.

Exploratory Data Analysis

We want to investigate the median number of goals scored per game between continents. The following figures show the distribution of goals per game scored for all teams and the median number of goals per game scored for each continent.

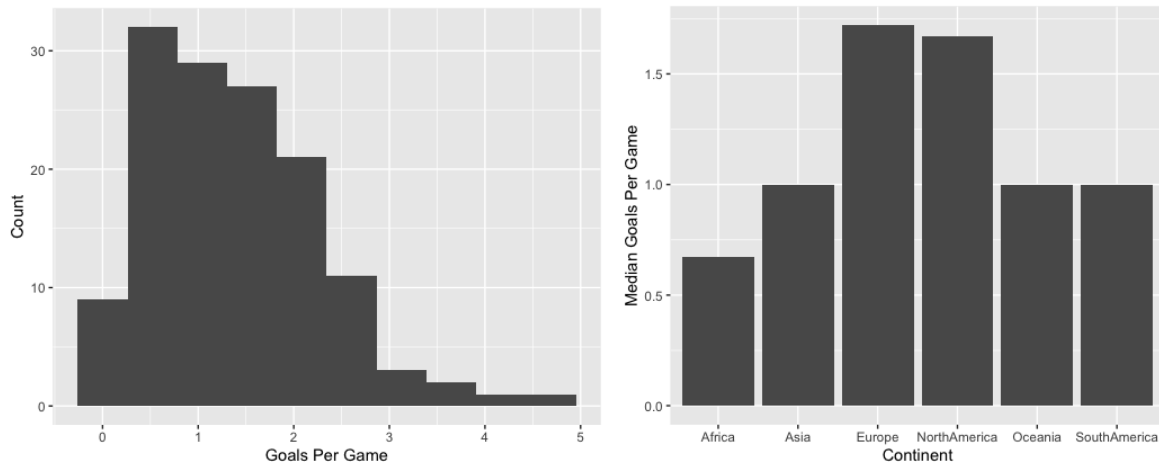


Figure 1. Figure 1a displays the distribution of goals per game for all observations. Figure 1b displays the median number of goals per game scored, broken down by continent.

Clearly, we can see that the number of goals per game scored is skewed right, so there are some high outliers that are affecting the shape of the data. Again, this is why the median would likely be a better measure of center than the mean. When looking at the data broken down by continent, we can see that the median number of goals appears to vary between continents, with Europe and North America having the highest values. However, in order to determine if there is a statistically significant difference between the medians, we must perform the Kruskal-Wallis test.

Conditions

When attempting to use the ANOVA test, there are three conditions that must be met: independence, normality, and constant variance. However, not all of these are satisfied with the Women's World Cup data, so that is why we must use the Kruskal-Wallis test instead. The independence condition is still met. Checking for independence is a little more difficult with this data set because it is data for all of the Women's World Cups, which essentially makes it a population. However, if we instead think of this as being a sample of all women's soccer in general, then we can assume independence. Unfortunately, this data does fail the normality condition. In order to check for normality, we must look at the distribution of goals per game for each continent group and the data set as a whole. From Figure 1a above, we can see that the data is heavily right skewed. Also, in Figure 2, seen below, most of the continents' distributions are also right skewed, at least slightly. This leads us to believe that the data may not be normally distributed, but to confirm this, we can use the Shapiro test. After completing this test in R, we see that the W-statistic is 0.95004 and the p-value is 0.000078. Since the p-value is less than 0.05, we conclude that the data is not normally distributed.

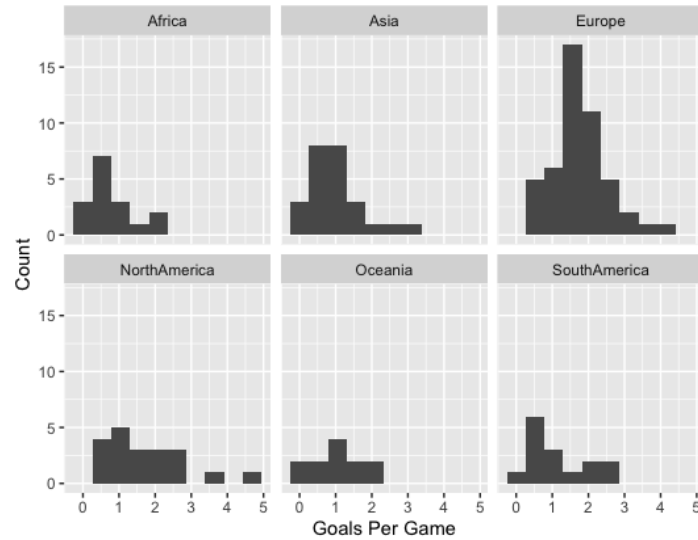


Figure 2. The distributions of goals per game broken down by continent.

The final condition to check is nearly constant variance within and between groups. Figure 3 below shows the side-by-side boxplots for goals per game for each continent. There is not a ton of variation between each of the plots, so that tells us that constant variance between groups is likely satisfied. To confirm, we can use the Levene test. After completing this test in R, we get that the F-statistic is 1.372 and the p-value is 0.239. Since 0.239 is greater than 0.05, we conclude that the constant variance condition is met. This is useful, but since normality failed, this still means that we cannot use the ANOVA test.

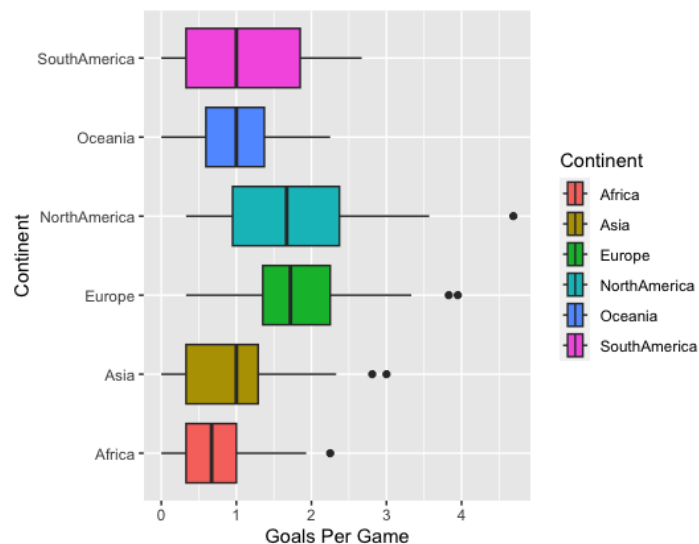


Figure 3. Side-by-side boxplots for goals per game, broken down by continent.

Therefore, all three conditions for the ANOVA test are not met, so we must continue with the Kruskal-Wallis test to compare the medians between groups.

Hypotheses

For our hypothesis test, we will test whether or not there is a difference in the median number of goals per game scored between continents.

H_0 : The median number of goals per game scored is the same between all continents.

H_A : At least one median is different.

We will test using a significance level of $\alpha = 0.05$ and use the Kruskal-Wallis test to complete the hypothesis test.

Hypothesis Test Results

After completing the Kruskal-Wallis test using R, we find the H-statistic to be 28.267 and the degrees of freedom is 5. Our test statistic leads to a p-value of 0.0000323, which is less than our significance level of $\alpha = 0.05$. This means that we reject the null hypothesis and conclude that the median number of goals per game scored is different for at least one continent.

Post-Hoc Comparison

Since we know that there is a difference in medians for at least one continent, we can now do another comparison to determine which medians are different from each other. We will use the Wilcoxon test with the Bonferroni correction, which compares each pair of medians using a lower significance level, which accounts for the inflated Type I error rate. The formula for our new significance level is $\alpha^* = \frac{\alpha}{K}$, where $K = \frac{k(k-1)}{2}$ and k is the number of groups. Since we have six continent groups, we get $K = 15$, so $\alpha^* = \frac{0.05}{15} = 0.003$. After doing the pairwise test in R, we get new p-values for each of the individual pairs of medians. The comparison between Europe and Africa has a p-value of 0.00087 and the comparison between Europe and Asia has a p-value of 0.00189, which are the only values that are lower than our new significance level.

Conclusion

After completing the post-hoc comparison, we can conclude that the median number of goals scored per game is different between Europe and Africa and between Europe and Asia, but there is no significant difference between any other continents.

Question 2

Which variables are the best predictors of winning the Women's World Cup?

Introduction

To determine which variables predict winning the World Cup, we first had to create a binomial response variable for winning. This was coded as 0 for a loss and 1 for a win. Since there are

only two levels for our response variable, this means that we should use a logistic regression model to predict the probability of winning.

Logistic Regression Model

When creating the logistic regression model, we tried several different variables, including goals per game, assists per game, number of matches played, continent, and average player age. These were tested individually and in combination with each other and the AIC criterion was compared between models to determine which was the best. The model with the lowest AIC value includes goals per game and average player age. The equation for this logistic regression model is

$$\log_e\left(\frac{p}{1-p}\right) = -30.7328 + 3.1510 * (\text{goals per game}) + 0.7979 * (\text{age})$$
 and the AIC value is 30.26.

These coefficients tell us that having a higher number of goals scored per game and having a higher average player age lead to a higher predicted probability of winning the World Cup.

Predictions

Using this model, we made predictions for the teams that have won the Women's World Cup in the past. We know which teams won for each year, and our dataset includes their average number of goals scored per game and the average age of their players. This will give us the probability of that team winning the World Cup, based on their statistics, but since we already know the actual outcome, this helps us to determine how accurate our model is. The results are summarized in Figure 4 below.

World Cup Year	Winner	Probability of Winning
1991	United States	0.9167
1995	Norway	0.7240
1999	United States	0.3953
2003	Germany	0.9629
2007	Germany	0.7120
2011	Japan	0.0082
2015	United States	0.1456
2019	United States	0.9706

Figure 4. Predicted probabilities of past World Cup winners winning the tournament based on our logistic regression model.

Out of the eight Women's World Cups, the model predicted fairly high chances of winning for five of the winners. However, the other three teams that actually won received less than a 20%

chance of winning. Based on the data, the teams that had higher probabilities of winning from the model were the same teams that had a higher number of average goals scored per game and had a higher average player age. This makes sense in relation to the coefficients in the equation, but it is still interesting that some teams had such low predicted probabilities.

Inference

Another method used to determine the accuracy of our model is looking at the statistical significance of each predictor variable. The model output in R gives slope coefficients for each variable and also a p-value. This allows us to create a hypothesis test for each predictor and decide which are significant. Assuming β_1 is the coefficient for goals per game and β_2 is the coefficient for age, the hypotheses are:

$H_0: \beta_1 = 0$, given that age is still included in the model

$H_0: \beta_2 = 0$, given that goals per game is still included in the model

The p-value for goals per game is 0.00052, which is less than the significance level $\alpha = 0.05$, so we can conclude that goals per game is a significant predictor in the model. The very low p-value means that it is very unlikely to get a slope coefficient of 3.1510 if there actually is not a relationship between goals per game and winning the World Cup, given age is included in the model. However, the p-value associated with age is 0.0679, which is greater than $\alpha = 0.05$, so we cannot reject the null hypothesis, and we get that age might not be a significant predictor. In order to determine if age should still be included in the model, the next step would be to make cross-validated predictions, but that is beyond the scope of this project.

Conclusion

Based on our logistic regression model, it appears that the number of goals scored per game and the average player age are the best predictors of whether or not a team will win the World Cup. A higher value for each of these variables increases a team's probability of winning, but goals per game is more significant than average player age.

Question 3

Which variables are the best predictors of how far a team makes it into the World Cup?

Introduction

This question is an extension of the logistic regression we just completed, however, instead of just looking at a binary response variable, like win or loss, we're looking at what stage of the tournament the team makes it to, which is a multi-level response variable. So, we must use a

slightly different kind of model, called ordinal logistic regression. This means that the response variable has more than two levels and there is a specific order to them. In this example, the levels go from Group Stage, to Knockout, to Final. When coding this variable, teams were assigned to the group stage if they only played three games, to the knockout stage if they played four or five games, and to the final stage if they were in the championship game (since the number of teams in the World Cup has changed over time and there is also a third place game, coding for this level was more complicated). Once we have this new kind of response variable, we can begin to create the model and make predictions, just like for logistic regression.

Ordinal Logistic Regression Model

When creating this model, we again tried many combinations of the variables from the data set, including goals per game, assists per game, average player age, and continent. Similarly to logistic regression, we compared the AIC criterion between models and the one with the lowest value again was goals per game and age, with a value of 180.0013. The slope coefficient for goals per game is 2.1162 and the slope coefficient for age is 0.3095, meaning that higher values for both variables increase a team's probability of making it to the next stage of the tournament. Since there are multiple levels to the response variable, there are multiple intercepts. The intercept for going from the group stage to the knockout stage is 10.0432 and the intercept for going from the knockout stage to the final is 14.1507, but these are more complicated to interpret in context.

Predictions

To test the accuracy of the model, we made some predictions with data from the 2019 Women's World Cup. We chose three teams, each of which made it to different stages of the tournament, and used our model to see their predicted probabilities of making it to each stage. The United States made it to the final, while France made it to the knockout stage, and Jamaica made it to the group stage. The results are summarized in Figure 5.

Country	Group Stage	Knockout	Final
United States	0.0016	0.0881	0.9103
France	0.0746	0.7560	0.1694
Jamaica	0.9053	0.0930	0.0017

Figure 5. Predicted probabilities of teams making it to each stage of the World Cup.

The model seems to predict very accurately what we already know to be true. The United States had a 91.03% chance of making it to the final, which is what they did. France had a 75.60% chance of making it to the knockout stage, which is where they ended up, and Jamaica had a 90.53% chance of only playing in the group stage, which is what happened. These results give

validity to our model because they agree fairly well with the observed data, which gives us confidence that the model is making accurate predictions.

Inference

Again, we can look at the statistical significance of each of the two predictor variables as another method of determining the accuracy of our model. Assuming that β_1 is the coefficient for goals per game and β_2 is the coefficient for age, the hypotheses for testing the predictors are:

$H_0: \beta_1 = 0$, given that age is still included in the model

$H_0: \beta_2 = 0$, given that goals per game is still included in the model

Then, the p-value for the goals per game coefficient is approximately zero, which is less than our significance level of $\alpha = 0.05$, so we can reject the null hypothesis and conclude that goals per game is a statistically significant predictor of how far a team makes it into the World Cup. The p-value for the age coefficient is 0.0053, which is also less than $\alpha = 0.05$, so we can again reject the null hypothesis and conclude that average player age is also a statistically significant predictor. This means that both of these variables should be included in the model.

Conclusion

Based on the ordinal logistic regression model we created, number of goals scored per game and average player age are the best predictors of how far a team will make it into the World Cup, and both are statistically significant in the model.

Citations

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